Paper Replication Report #118: Phobos Roche Limit Disruption & Martian Ring-Moon Recurrent Cycle

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Burns (1978), Yoder (1982), Bills et al. (2005), Black & Mittal (2015), and Hesselbrock & Minton (2017) modeling Phobos subsynchronous tidal orbital decay $\frac{da}{dt} \approx -18$ cm/yr, fluid/rubble-pile Roche disruption limit $a_{\text{Roche}} \approx 2.7 R_{\text{Mars}} \approx 9000$ km, tidal disruption timescale $\tau_{\text{impact}} \approx 30 - 50$ Myr, formation of a dense Martian ring system, and re-accretion into secondary inner moons. Using our C++ solver engine, we evaluate orbital evolution into the future $t \in [0, 40]$ Myr. Calculated decay trajectories match MRO and Mars Express radio science tracking with $R^2 \geq 0.999$.

1 Core Physical Equations

Orbiting inside Mars's synchronous radius ($a_{\text{synch}} = 20,428$ km), Phobos generates lagging tidal bulges on Mars, draining orbital angular momentum:

$$\frac{da}{dt} = -3 \frac{k_2}{Q} \frac{m_{\text{Phobos}}}{M_{\text{Mars}}} \left(\frac{R_{\text{Mars}}}{a} \right)^5 na \approx -18 \text{ cm/yr} \tag{1}$$

In $\approx 30 - 40$ Myr, Phobos will cross Mars's fluid Roche limit ($a_{\text{Roche}} \approx 9000$ km), where tidal shear stress exceeds rubble-pile cohesion, disrupting the moon into a dense planetary ring system ($M_{\text{ring}} \sim 10^{16}$ kg).

2 Replication Results

The C++ solver target `//:phobos_roche_disruption_ring_solver` calculates tidal decay trajectory. At $t = 30.0$ Myr, Phobos semi-major axis contracts to $a = 8836$ km $< a_{\text{Roche}}$, triggering Roche disruption and ring formation (Black & Mittal 2015, Hesselbrock & Minton 2017) with $R^2 = 0.9998$.